Aldrick Victor Tadeo Arellano

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Education

Bachelors of Science, Robotics and Digital Systems Engineering 2022 — 2026
Monterrey Institute of Technology and Higher Education (ITESM) *GPA: 4.0/4.0*
- Academic Excellence Scholarship recipient.

Technical Degree, Mechatronics 2019 — 2022
CBTis #118 Technical High School *GPA: 4.0/4.0*

Skills

Programming:
- Python
- C/C++
- C#
- JavaScript
- MATLAB
- VHDL

Technologies:
- ROS2
- Linux
- Git/GitHub
- Docker
- Machine Learning
- Computer Vision

CAD/CAE:
- SolidWorks
- Autodesk Fusion
- KiCAD
- Siemens NX
- AutoCAD

Languages:
- English (Native)
- Spanish (Native)
- Japanese (Basic)

Soft Skills:
- Critical Thinking
- Adaptability
- Leadership
- Time Management

Methodologies:
- Agile/Scrum

Work Experience

Software Engineer Intern, *GE Aerospace* Nov 2025 — Present
- Designed a usage-telemetry framework for multiple internal CAD/CAE (Siemens NX) and PLM/data tools; defined event model + PostgreSQL schema for hundreds of daily users.
- Integrated telemetry into engineering toolchains via a C#/.NET service wrapper, sending events through REST + internal service calls.
- Developed NX Open plugins to employ NX workflows with internal tools and capture usage/error data with minimal impact to engineers.
- Built Grafana dashboards to track adoption, workflow bottlenecks, and reliability trends across CAD/PLM tooling.

Telemetry Development Division Lead & Founder, *Electrum Performance Racing Team* Jun 2024 — Oct 2025
- Founded and led the Telemetry Division, building a robust real-time data acquisition system designed for high-stress environments.
- Developed failsafe telemetry solutions using serial communication protocols, reducing transmission failures by 15%.
- Improved cross-team collaboration by implementing Fusion 360′s cloud-based version control, cutting design iteration times by 10%.
- Directed a cost optimization initiative, reducing manufacturing and component expenses by 35% while maintaining system reliability.
- Engineered firmware components to streamline workflows and enhance data accuracy across multiple vehicle subsystems.

Projects

Robotic Design & Software Lead, IEEE Robosoft Competition 2026 Aug 2025 — Present
- Designed a soft robotic gripper for strawberry harvesting, including the tool, control software, and circuitry; optimized through iterative

design to reduce material costs and enhance flexibility.
- Developed IoT-based crop monitoring hardware and software for greenhouse environments, improving real-time tracking of crop health.
- Built an AI vision system for in-field strawberry classification, optimized for deployment on both remote servers and edge devices.
- Created an object localization system using an in-hand RGB-D camera to guide robotic tool positioning with precision

AI Navigation Systems Developer, Manchester Robotics & Tec de Monterrey Aug 2025 — Present
- Developed autonomous navigation and control systems on the MCR2 Puzzlebot platform, enabling robust mobile robotics operation.
- Implemented traffic sign and traffic light recognition systems, improving reliability, accuracy, and environmental adaptability by 33%.
- Enhanced navigation and positioning precision by fusing visual and internal sensors for improved real-world performance.
- Built vision redundancy features, reducing false positives and lowering detection/decision error rates by 15%.
- Redesigned the ROS2 node-topic architecture, optimizing data throughput and minimizing system failure points.

Certificates

Fundamentals of Deep Learning, NVIDIA Credential ID: 1tO0Ys3ITkGJkXM3sgBKrQ
Generative AI with Diffusion Models, NVIDIA Credential ID: TauXuWfURMOBYNutOVkopw
OpenCV Bootcamp, OpenCV University Credential ID: 65a25e083f50497dba5f5538026087
UR e-Series Tracks (Core, Pro, Application), Universal Robots